SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High) Assessment
Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**

1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are	BL1 – Remember	5
	characterized by a unique schedule Number, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5

3	University Registrar’ s Office Management System A university registrar’ s office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar’ s office.	BL2 – Understand	5
4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient (SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5

5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. ● A department controls a number of PROJECT. Projects have a unique name, number and a single location. ● Company' s EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. ● Employee' s DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5
8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I NS avesh hussain(192525241) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understandin g
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandabl e with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. UPS Product Tracking Information System – ER Diagram

Entities

Shipped_Item

- Item No (PK)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

Retail_Center

- Center_ID (PK)
- Type
- Address

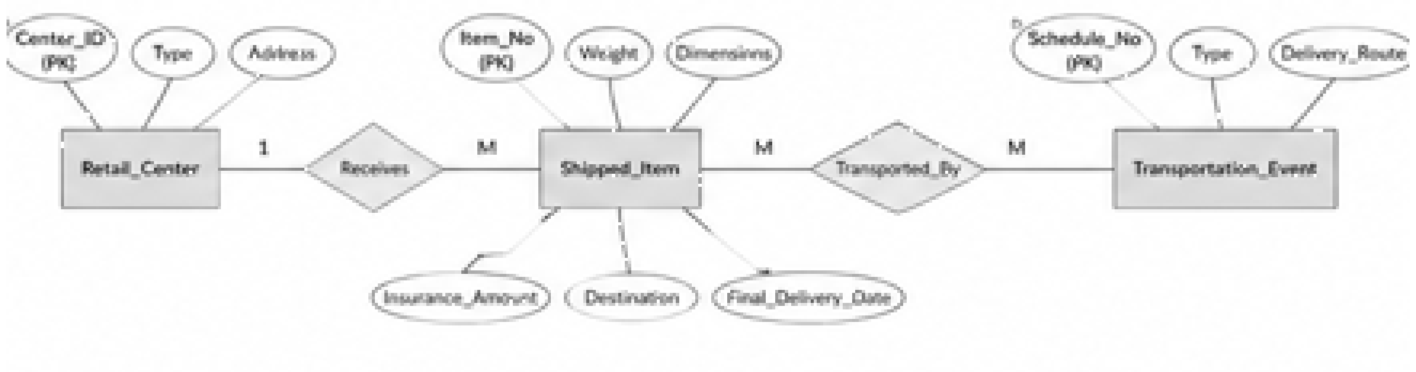
Transportation_Event

- Schedule_No (PK)
- Type (Flight/Truck)
- Delivery_Route

Relationships

- Retail_Center **receives** Shipped_Item (1:M)
- Shipped_Item **transported through** Transportation_Event (M:N)

ER Diagram



2. Course Management System – ER Diagram

Entities

Course

- Name (PK)
- Department
- Course_No

Section (Weak Entity)

- Section_No (Partial Key)
- Enrollment

Room

- Room_No (PK)
- Capacity
- Building

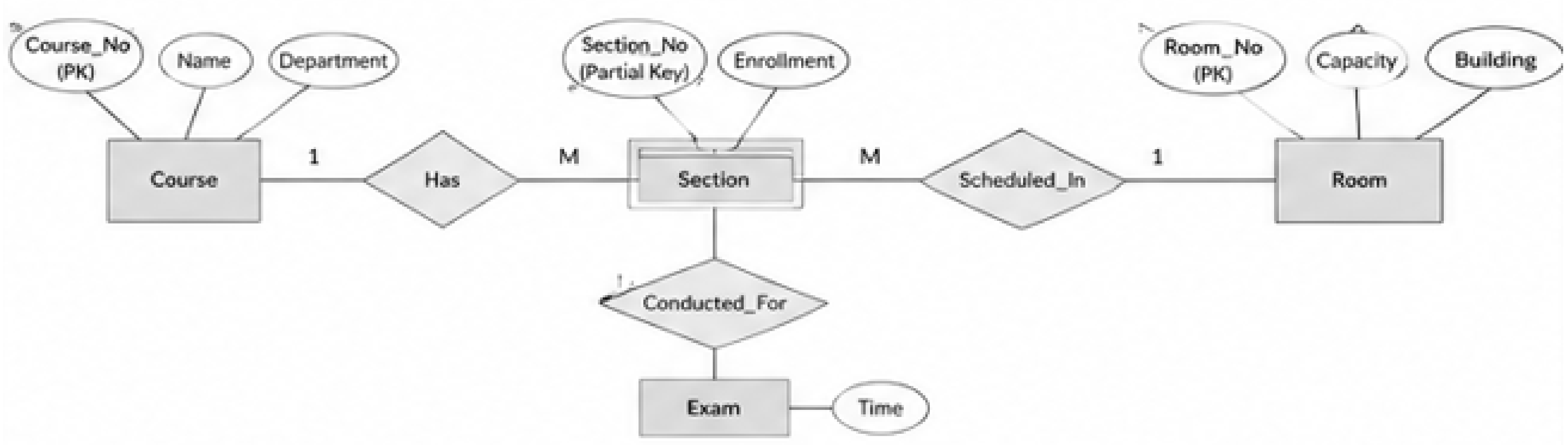
Exam

- Time

Relationships

- Course **has** Section (1:M)
- Section **scheduled in** Room
- Section **has** Exam

ER Diagram



3. University Registrar's Office Management System – ER Diagram

Entities

Course

- Course_No (PK)
- Title
- Credits
- Syllabus
- Prerequisites

Course_Offering

- Year
- Semester
- Section_No
- Timings
- Classroom

Student

- Student_ID (PK)
- Name
- Program

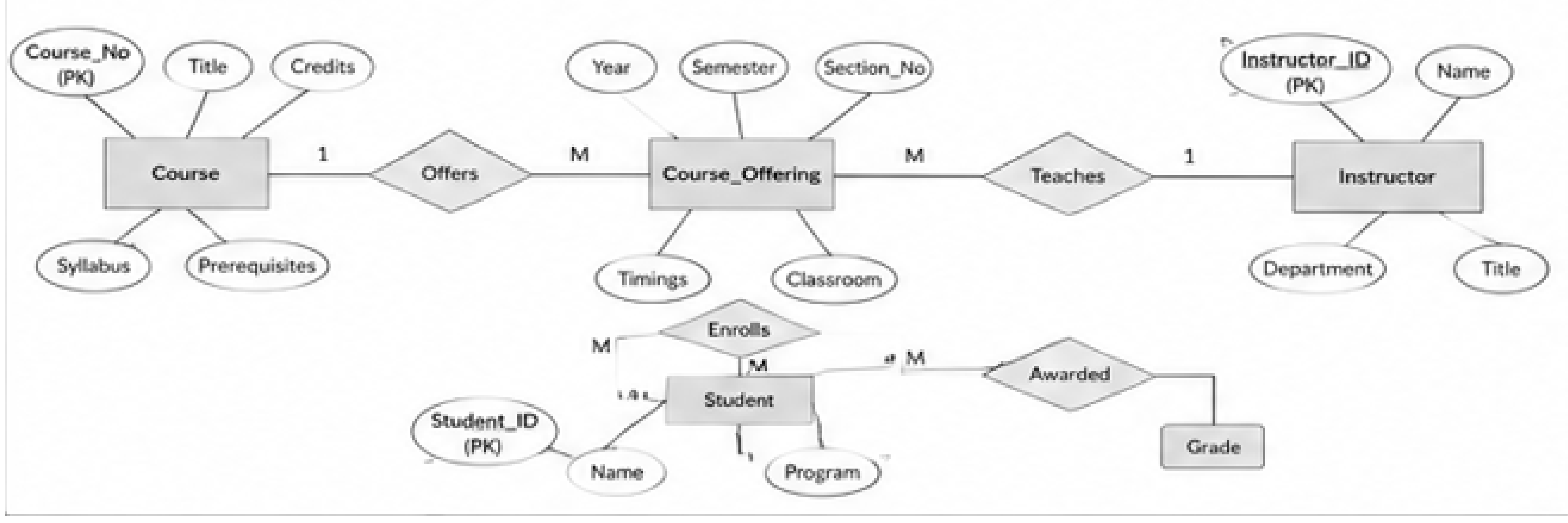
Instructor

- Instructor_ID (PK)
- Name
- Department
- Title

Relationships

- Course **offers** Course_Offering
- Instructor **teaches** Course_Offering
- Student **enrolls in** Course_Offering
- Enrollment stores **Grade**

ER Diagram



4. Hospital Management System – ER Diagram and Tables

Entities

Patient

- SS# (PK)
- Name
- Insurance

Physician

- Name (PK)
- Specialization

Test_Log

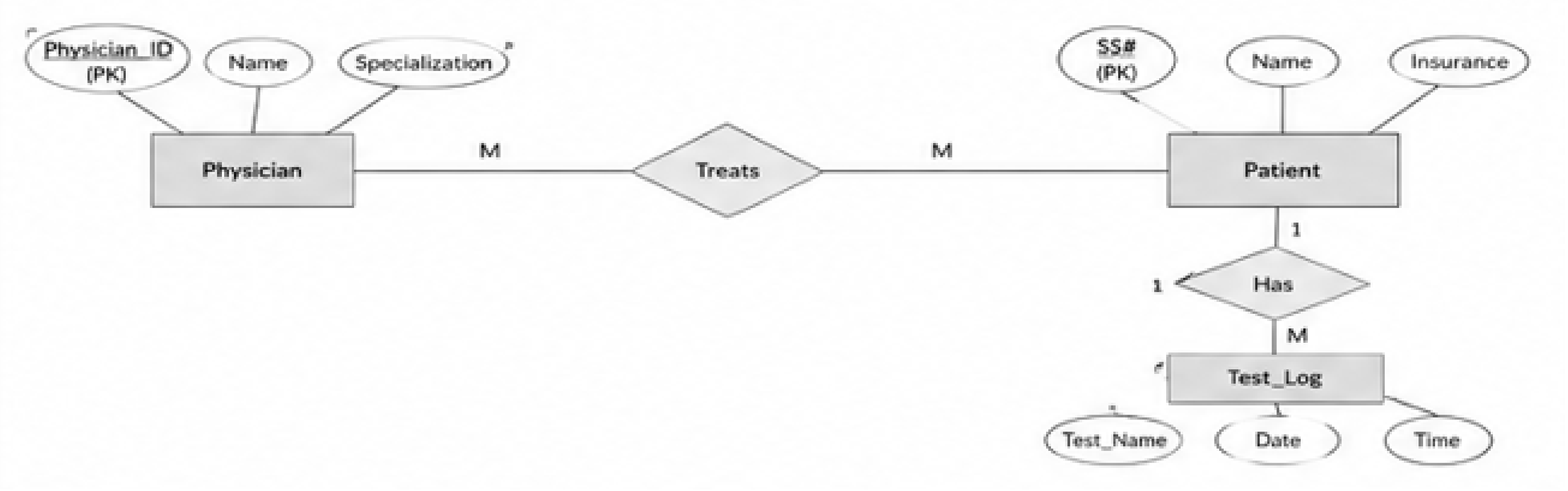
- SS#
- Test_Name

- Date
- Time

Relationships

- Physician **treats** Patient
- Patient **has** Test_Log
-

Tables



5. Company Project Management System – ER Diagram

Entities

Department

- Dept_ID (PK)
- Dept_Name

Employee

- SSN (PK)
- Name
- Address
- Salary
- Sex
- Birth_Date

Project

- Project_No (PK)
- Project_Name
- Location

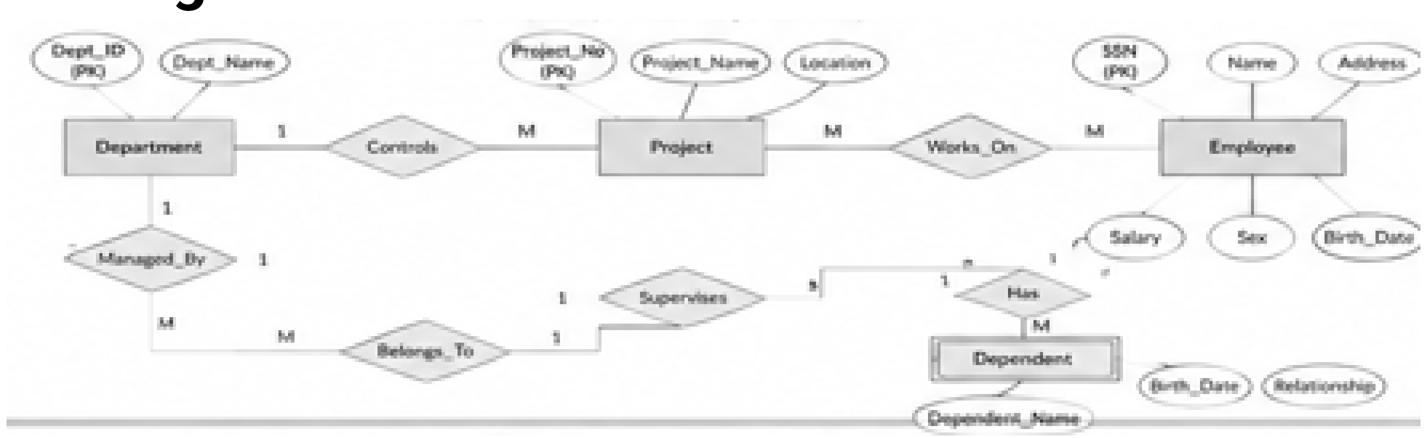
Dependent

- Dependent_Name
- Birth_Date
- Relationship

Relationships

- Department **managed by** Employee (Manager_Start_Date)
- Department **controls** Project
- Employee **works on** Project (Hours/Week)
- Employee **belongs to** Department
- Employee **supervises** Employee
- Employee **has** Dependent

ER Diagram



6. Schema and Instance

Schema

A **schema** is the overall structure or blueprint of a database.

Example

Student(Student_ID, Name, Department)

Instance

An **instance** is the actual data stored in the database at a particular time.

Student_ID	Name	Department
101	Rahul	CSE
102	Priya	AI

Difference

Logical Schema	Physical Schema	View Schema
Defines tables and relationships	Defines storage details	User-specific view
Used by designers	Used by DBMS	Used by users
Independent of storage	Deals with files/indexes	Shows selected data

7. Bank Management System – ER Diagram

Entities

Customer

- Customer_ID (PK)
- Name
- Address
- Phone

Account

- Account_No (PK)
- Type
- Balance

Branch

- Branch_ID (PK)
- Branch_Name

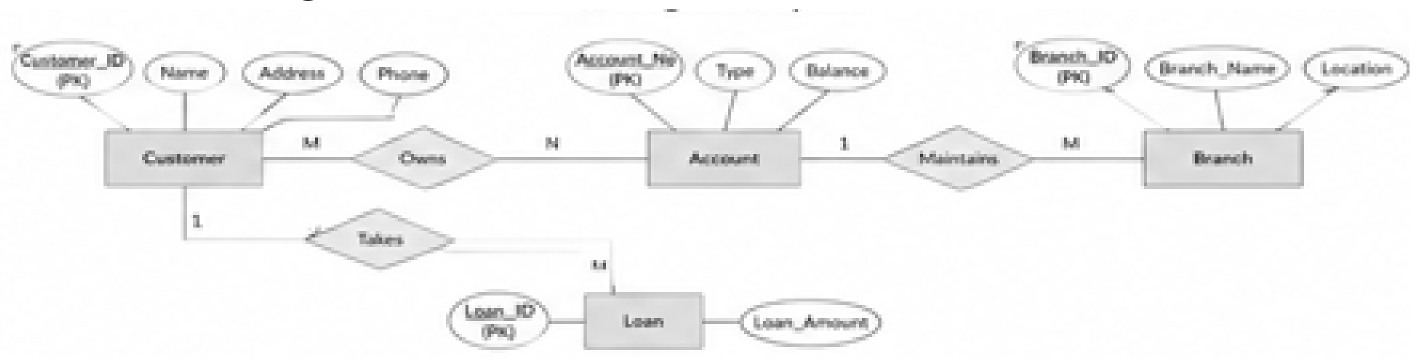
Loan

- Loan_ID (PK)
- Amount

Relationships

- Customer **owns** Account
- Branch **maintains** Account
- Customer **takes** Loan

Text ER Diagram



8. Extended ER (EER) Model

The **Extended Entity Relationship (EER) Model** is an extension of the ER model that supports advanced database concepts like inheritance and specialization.

Generalization

Combines similar lower-level entities into a higher-level entity.

Example

Car

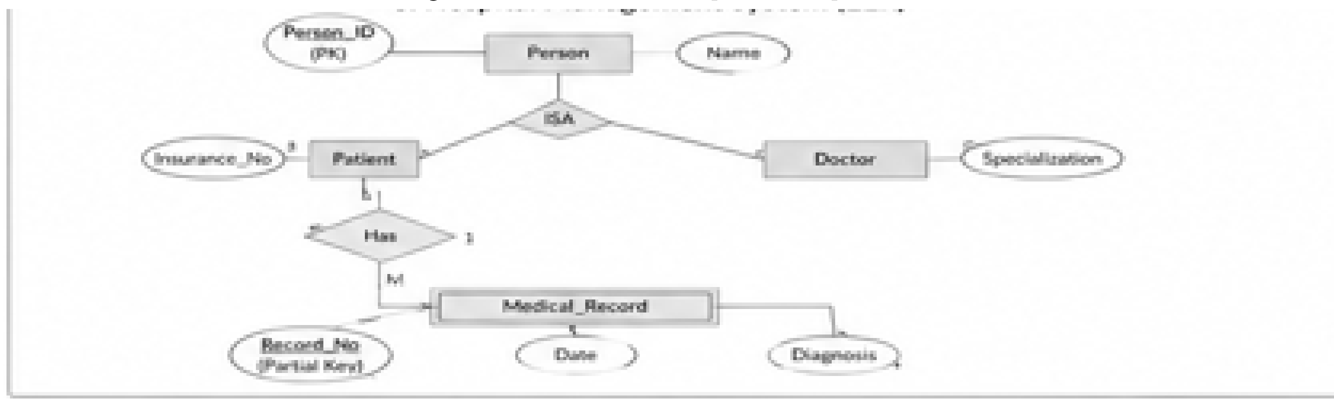
Bike

↓

Vehicle

Specialization

Divides one entity into multiple specialized entities.



9. EER Diagram for Hospital Management System

Entities

Person

- ID
- Name
- Address

Specialization

```
      Person
     /    \
  Patient   Doctor
```

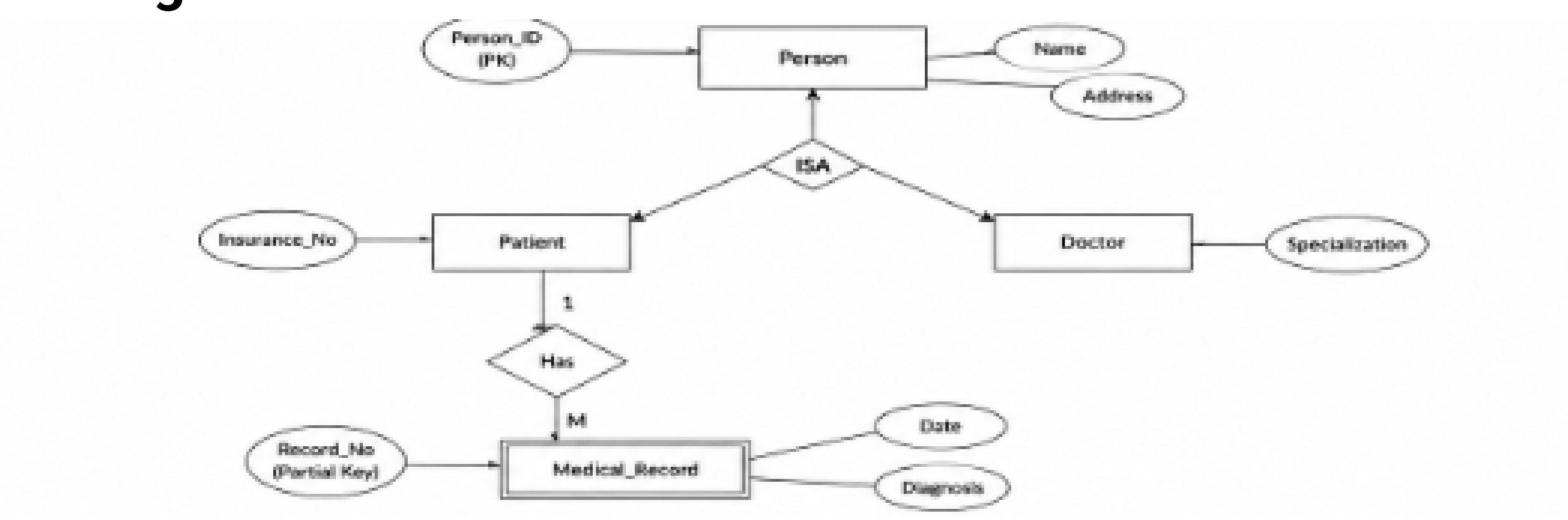
Weak Entity

Medical_Record

- Record_No (Partial Key)
- Date
- Diagnosis

Depends on **Patient**.

ER Diagram



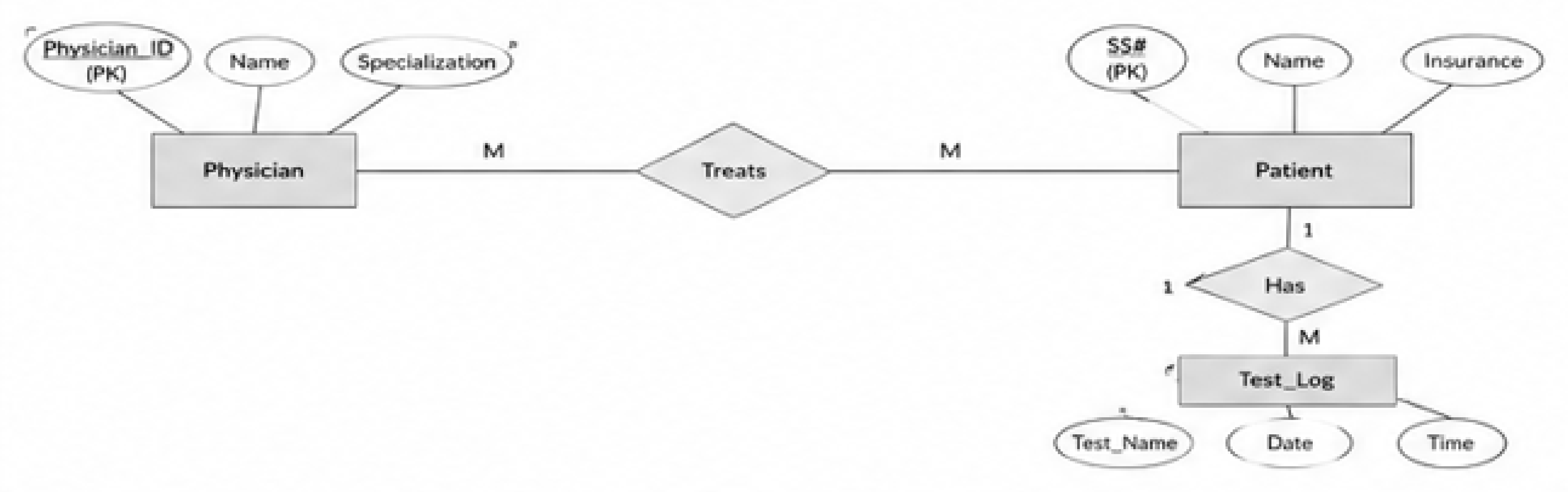
10. Role of DBA (Database Administrator)

A **Database Administrator (DBA)** is responsible for managing, maintaining, and securing a database.

Responsibilities

1. Install and configure the DBMS.
2. Create and maintain databases.
3. Manage user accounts and access permissions.
4. Perform backup and recovery of data.
5. Ensure database security.
6. Monitor database performance.
7. Optimize queries and indexing.
8. Maintain data integrity and consistency.
9. Troubleshoot database issues.
10. Plan storage and future database growth.

ER Diagram



Conclusion:

The DBA ensures that the database is secure, reliable, efficient, and always available for users.